

A dimension-free lower bound for total coherence of Parseval frames

Full answer to an open question in arXiv:1910.01733

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Status. The source question has an affirmative answer over both $\mathbb{R}$ and $\mathbb{C}$. A Haar-random Parseval frame already has the requested order of total coherence in expectation. In particular, one may take the universal constant

$$c = \frac{2\sqrt{2}}{\pi^{3/2}} = 0.5079\dots$$

Source question

For an M -vector Parseval frame $\Phi = (\varphi_1, \dots, \varphi_M)$ in $\mathbb{F}^N$, the source defines

$$\text{TC}(\Phi) = \sum_{i \neq j} |\langle \varphi_i, \varphi_j \rangle|.$$

It asks whether some $c > 0$, independent of N, M , always permits a Parseval frame satisfying

$$\text{TC}(\Phi) \geq c\sqrt{N(M-N)(M-1)} \tag{Q}$$

[1, p. 25]. The next paragraph explicitly proposes calculating the expectation for a random Parseval frame.

Another type of question that we can consider is that of getting better bounds on the optimal values of the functions that we are optimizing. For example, in Proposition 2.3 we showed that $\text{TC}(\Phi) \geq \max\{N, M-N\}$ for any equal norm frame, but if Φ is a solution to (2.2) then $\text{TC}(\Phi)$ is most likely quite a bit bigger than that, so it is desirable to have a tighter lower bound. For example, one natural question is does there exist a constant $c > 0$ independent of N and M such that there always exists Parseval frame Φ with

$$\text{TC}(\Phi) \geq c\sqrt{N(M-N)(M-1)}?$$

We can ask a similar question about $V_k(\Phi)$ with the upper bound given in Proposition 3.2 (and (5.6), which is the same bound). Of course it is also desirable to have tighter upper bounds in the cases where the bounds we have cannot be saturated. For the case of $NE_k(\Phi)$ we don't even have a good upper bound since the bound we get from (4.2) can never be saturated so a bound in either direction is desirable.

One way to approach finding bounds on the optimal values of these functions is to try to calculate the expected value of them for a random Parseval frame or a random equal norm frame. It is clear what is meant by a random equal norm frame. For a random Parseval frame recall the correspondence between unitary equivalence classes of Parseval frames in $\mathcal{P}(M, N)$

Source evidence: arXiv:1910.01733, PDF page 25.

Answer and exact expectations

Assume $1 \leq N < M$, and put

$$q = M - N, \quad D = M - 1.$$

Choose a Haar-uniform N -dimensional subspace of $\mathbb{F}^M$, and let P be the orthogonal projection onto it. Choose any Parseval frame Φ whose Gram matrix is P . (Equivalently, take an $N \times M$ matrix with orthonormal rows and Gram matrix P .)

Theorem 1. For the random Parseval frame above, if $\mathbb{F} = \mathbb{C}$, then

$$\mathbb{E} \text{TC}(\Phi) = D \frac{\Gamma(N + \frac{1}{2})}{\Gamma(N)} \frac{\Gamma(q + \frac{1}{2})}{\Gamma(q)} \frac{\Gamma(\frac{3}{2})\Gamma(D)}{\Gamma(D + \frac{1}{2})} \geq \frac{\pi^{3/2}}{8} \sqrt{NqD}. \quad (1)$$

If $\mathbb{F} = \mathbb{R}$, then

$$\mathbb{E} \text{TC}(\Phi) = 2D \frac{\Gamma(\frac{N+1}{2})}{\Gamma(\frac{N}{2})} \frac{\Gamma(\frac{q+1}{2})}{\Gamma(\frac{q}{2})} \frac{\Gamma(\frac{D}{2})}{\sqrt{\pi} \Gamma(\frac{D+1}{2})} \geq \frac{2\sqrt{2}}{\pi^{3/2}} \sqrt{NqD}. \quad (2)$$

Consequently, in either field there exists a Parseval frame satisfying (Q) with $c = 2\sqrt{2}/\pi^{3/2}$.

Proof intuition. The Gram matrix of a Parseval frame is a projection. For a random projection, all off-diagonal entries have the same distribution, so it suffices to calculate one of them. Once its first diagonal entry d is fixed, the off-diagonal part of the first column has forced length $\sqrt{d(1-d)}$ and a uniform spherical direction. The length and one spherical coordinate have elementary beta moments. Multiplying those moments gives (1) and (2); simple gamma-ratio bounds then produce constants independent of the dimensions.

Proof

Write $d = P_{11}$. Since $P^2 = P$,

$$\sum_{j=2}^M |P_{j1}|^2 = (P^2)_{11} - P_{11}^2 = d(1-d). \quad (3)$$

The subgroup of the orthogonal or unitary group that fixes the first coordinate preserves the law of P and acts transitively on the unit sphere of $\mathbb{F}^D$. Conditional on d , therefore,

$$(P_{21}, \dots, P_{M1}) \stackrel{d}{=} \sqrt{d(1-d)} Z, \quad (4)$$

where Z is uniform on that sphere; its conditional law is the same for every d . Thus

$$\mathbb{E}|P_{12}| = \mathbb{E}\sqrt{d(1-d)} \mathbb{E}|Z_1|. \quad (5)$$

These distributions can be seen directly from normalized Gaussian vectors. The first row of a Haar orthogonal/unitary matrix is uniform on the sphere, and d is the sum of the squared moduli of its first N coordinates. Hence

	d	$ Z_1 ^2$
$\mathbb{F} = \mathbb{C}$	$\text{Beta}(N, q)$	$\text{Beta}(1, D-1)$
$\mathbb{F} = \mathbb{R}$	$\text{Beta}(N/2, q/2)$	$\text{Beta}(1/2, (D-1)/2)$

For $D = 1$, simply $|Z_1| = 1$, which agrees with the gamma formulas below. Using

$$\mathbb{E}\sqrt{X(1-X)} = \frac{B(a + \frac{1}{2}, b + \frac{1}{2})}{B(a, b)} \quad \text{for } X \sim \text{Beta}(a, b),$$

and the analogous beta moment for $|Z_1|$, we obtain

$$\begin{aligned} \mathbb{F} = \mathbb{C} : \quad \mathbb{E}\sqrt{d(1-d)} &= \frac{1}{M} \frac{\Gamma(N + \frac{1}{2})}{\Gamma(N)} \frac{\Gamma(q + \frac{1}{2})}{\Gamma(q)}, \\ \mathbb{E}|Z_1| &= \frac{\Gamma(\frac{3}{2})\Gamma(D)}{\Gamma(D + \frac{1}{2})}. \end{aligned} \quad (6)$$

$$\begin{aligned} \mathbb{F} = \mathbb{R} : \quad \mathbb{E}\sqrt{d(1-d)} &= \frac{2}{M} \frac{\Gamma(\frac{N+1}{2})}{\Gamma(\frac{N}{2})} \frac{\Gamma(\frac{q+1}{2})}{\Gamma(\frac{q}{2})}, \\ \mathbb{E}|Z_1| &= \frac{\Gamma(\frac{D}{2})}{\sqrt{\pi} \Gamma(\frac{D+1}{2})}. \end{aligned} \quad (7)$$

By symmetry and the ordered-pair convention in the definition of TC,

$$\mathbb{E} \text{TC}(\Phi) = M(M-1) \mathbb{E}|P_{12}|.$$

Combining this identity with (5)–(7) proves the exact formulas in the theorem.

It remains to give dimension-free lower bounds. For integers $n \geq 1$, put

$$b_n = \frac{\Gamma(n + \frac{1}{2})}{\Gamma(n)}.$$

Since $b_1 = \sqrt{\pi}/2$ and $b_{n+1}/b_n = (n + \frac{1}{2})/n$, induction gives

$$b_n \geq \frac{\sqrt{\pi}}{2} \sqrt{n}. \quad (8)$$

Indeed, the inductive inequality after squaring reduces to $(n + \frac{1}{2})^2 \geq n(n + 1)$. Log-convexity of Γ gives

$$\Gamma(D + \frac{1}{2})^2 \leq \Gamma(D)\Gamma(D + 1) = D\Gamma(D)^2. \quad (9)$$

Substituting (8)–(9) and $\Gamma(3/2) = \sqrt{\pi}/2$ in (1) gives its stated complex lower bound.

For the real case put

$$a_n = \frac{\Gamma(\frac{n+1}{2})}{\Gamma(\frac{n}{2})}.$$

The bases $a_1 = 1/\sqrt{\pi}$ and $a_2 = \sqrt{\pi}/2 \geq \sqrt{2/\pi}$, together with $a_{n+2} = ((n + 1)/n)a_n$, prove by induction that

$$a_n \geq \sqrt{n/\pi}. \quad (10)$$

Here the inductive inequality after squaring is just $(n + 1)^2 \geq n(n + 2)$. A second use of log-convexity gives

$$\frac{\Gamma(\frac{D+1}{2})}{\Gamma(\frac{D}{2})} \leq \sqrt{D/2}. \quad (11)$$

Equations (10)–(11) give the real lower bound in (2). Since $\pi^{3/2}/8 > 2\sqrt{2}/\pi^{3/2}$, the real constant works in both fields. Finally, a nonnegative random variable has some realization at least as large as its expectation. That realization supplies the desired frame and completes the proof. $\square$

Scope and provenance

This proves precisely the source’s dimension-free existential lower bound; it does not assert that a random frame is an optimizer or answer the separate structural questions in the same Discussion. Exact arXiv-id, title, quoted-question, and core-keyword searches through 13 August 2026 found no later arXiv theorem settling this question. The proof is a new result of this run. All probability and gamma estimates needed for the conclusion are derived above.

References

- [1] J. Cahill and P. G. Casazza, *Optimal Parseval frames: Total coherence and total volume*, arXiv:1910.01733 (2019); Linear and Multilinear Algebra 71 (2023), 712–727. See the open question and proposed random-frame route on PDF page 25.